
Operational Amplifiers

Exercise Book

IESE Exercise Book 2 of 3

Problems only — attempt them first; answers (final results, no worked steps) are at the back. Use E12 values and quote realised errors.

How to use this book

This is the problem set distilled from the *Operational Amplifiers* solving manual. It contains **only exercises and their answers** — no theory and no worked derivations. Each chapter mirrors a chapter of the manual, so when you are stuck on a *method*, open the manual at the same chapter and read its Protocol and Worked Example.

Key Idea

Component-design exercises have more than one valid answer: any resistor pair with the right *ratio* works. The key gives one E12 choice and the realised value; if yours differs but the ratio and realised error match, you are right.

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Chapter 1

Circuits with the ideal Operational Amplifier

Exercise 1.1 Non-inverting design, $G = 3$

Design a non-inverting voltage amplifier of gain $A_V = 3$. Pick $R_1 = 12\text{ k}\Omega$, find R_2 , give the nearest E12 value, and state the realised gain and its relative error.

Exercise 1.2 The five amplifiers

(a) A non-inverting stage of $G = 10000$ on an op-amp with $A_d = 10^5$: find β and the loop gain $T = A_d\beta$, and comment. (b) Is a non-inverting stage of $G = 0.707$ possible? (c) Give the resistor ratio R_2/R_1 for inverting gains -3 , -1 , -0.707 .

Exercise 1.3 Transresistance amplifier

A current-to-voltage converter must map $I_S \in [0, 20\text{ mA}]$ to $V_{out} \in [0, -5\text{ V}]$. Find R_2 and state what the op-amp must be able to do.

Chapter 2

The real Operational Amplifier

Exercise 2.1 Does higher gain hurt?

An op-amp has $A_d \in [100\text{k}, 500\text{k}]$. For a closed-loop gain $A_V = 1000$, find the realised-gain spread and the relative uncertainty ε_{A_V} .

Exercise 2.2 Closed-loop impedances of a follower

An LM741 ($r_d = 2\text{ M}\Omega$, $r_o = 75\ \Omega$, $A_d = 2 \times 10^5$) is wired as a unity-gain follower. Find R_{in} and R_{out} .

Exercise 2.3 Lower bound on R_2 from I_{MAX}

$V_{PS} = 15\text{ V}$, $I_{load} = 20\text{ mA}$, non-inverting $A_V = 10$. (a) Bound R_2 and give an E12 resistor pair. (b) What does the bound become if the load only ever draws 2 mA?

Exercise 2.4 CMRR figure

A precision op-amp lists $\text{CMRR}_{dB} = 120\text{ dB}$ and $A_d = 10^6$. Find the common-mode gain A_C .

Exercise 2.5 Output swing into a load

$I_{MAX} = 25\text{ mA}$, $V_{MAX} = 10\text{ V}$. Find the peak swing into (a) $R_L = 200\ \Omega$ and (b) $R_L = 1\text{ k}\Omega$, and (c) the crossover load.

Exercise 2.6 A full R_2 window

$A_V = 20$, $V_{OFF} = 2\text{ mV}$, $I_{OFF} = 20\text{ nA}$, $V_{PS} = 12\text{ V}$, $I_{load} = 10\text{ mA}$. (a) Give the window for R_2 and pick a value. (b) With $R_2 = 100\text{ k}\Omega$ and $R_3 = R_1 \parallel R_2$, estimate the residual output offset.

Chapter 3

Frequency response, stability and bandwidth

Running device for this chapter: $A_{d0} = 100$ dB, $f_{p1} = 20$ kHz, $f_{p2} = 1$ MHz.

Exercise 3.1 Just stable

With no compensation: (a) at which closed-loop gain is the stage *just* stable ($\phi_M = 45^\circ$)? (b) Check $A_V = 5000$ explicitly — stable or not?

Exercise 3.2 Method 1 on a follower

The device is used as a *voltage follower* and stabilised by Method 1 (lower A_d). Find the required attenuation M , the compensated A_{d0} , and the frequency where $|T|$ now crosses 0 dB.

Exercise 3.3 Method 2.1 at $A_V = 100$

Using an external compensation capacitor (Method 2.1) at $A_V = 100$: where must the moved first pole f_{p1}^{new} be placed, and what DC loop gain survives?

Exercise 3.4 GBWP: pass or fail?

A non-inverting stage needs $A_{V0} = 25$ and a bandwidth $f_C = 50$ kHz from a part with GBWP = 1 MHz. Does it meet spec? Give two fixes.

Exercise 3.5 Slew rate

An LM741 ($SR = 0.5$ V/ μ s) must output a 10 kHz sine. (a) Largest amplitude V_p free of slewing? (b) If a 10 V peak is required, above what frequency does it slew?

Chapter 4

Applications

Exercise 4.1 Generalised adder/subtractor

Realise (a) $v_{out} = -2v_1 - 3v_2 - 4v_3$ and (b) the difference $v_{out} = 6(v'_1 - v''_1)$. Give the resistors (in terms of R_f) and the balancing element.

Exercise 4.2 Integrator response

A practical integrator has $RC = 0.25$ ms. (a) Feed it a 100 mV sine at 500 Hz: output amplitude and phase? (b) Remove the DC-stabilising R_2 and take $I_{BIAS} = 100$ nA: starting from 0, how long until the output reaches a 12 V rail?

Exercise 4.3 Square wave to triangle

An integrator turns a ± 1 V, 1 kHz square wave into a 2 V peak-to-peak triangle. Find the required RC , then pick C and R .

Chapter 5

The Operational Amplifier as a comparator

Exercise 5.1 Bare comparator with noise

Build an inverting comparator on a single 5 V supply. (a) Choose an E12 divider from the rail giving $V_{REF} = 2.0$ V. (b) State the output rule. (c) With noise $N_A = 50$ mV, give the input band where the output is undefined and the minimum hysteresis a Schmitt replacement would need.

Exercise 5.2 Symmetric inverting Schmitt

Design an inverting Schmitt trigger with $V_{OH} = +12$, $V_{OL} = -12$ V and thresholds at ± 2 V. (a) Find V_{REF} and R_2/R_1 and pick E12 values. (b) What is the largest noise amplitude it tolerates?

Exercise 5.3 Non-inverting Schmitt + source loading

Design a non-inverting Schmitt trigger with $V_{OH,OL} = \pm 12$ V and thresholds at ± 3 V. (a) Find V_{REF} and an E12 pair R_1, R_2 ; give the actual thresholds. (b) The source has $R_S = 1$ k Ω : what do the thresholds become, and what is the lesson?

Exercise 5.4 Symmetric astable

Design a 10 kHz astable with $V_{OH} = +12$, $V_{OL} = -12$ V and symmetric thresholds at half the output levels, $V_{S1,2} = \pm 6$ V. Give R_1, R_2 and R, C .

Exercise 5.5 Triangle/square generator

Design a triangle/square generator for a 1 kHz, 5 V peak-to-peak triangle with $V_{OH}, V_{OL} = \pm 10$ V. Give R_1, R_2 (E12) and R, C .

Chapter 6

Laboratory: Op-amp characterisation

Exercise 6.1 Bandwidth- or slew-limited step?

On the bench an LM741 follower is driven with a small step and then a large step. The small step settles with a rise time set by $0.35/BW$; the large step ramps at a constant slope. Which mechanism limits each, and which rise time grows with the step height?

Chapter 7

Answers

Final results only. Design exercises quote one valid E12 choice — any pair with the same ratio is correct.

Chapter 1 — Circuits with the ideal Operational Amplifier

- 1.1** $R_2 = 24\text{ k}\Omega$; nearest E12 $22\text{ k}\Omega$ gives $G' = 1 + 22/12 = 2.83$, relative error 5.6 %.
- 1.2** (a) $\beta = 10^{-4}$, $T = A_d\beta = 10$ — small, so feedback benefits are weak. (b) No: a non-inverting stage cannot give $|A_V| < 1$. (c) Ratios 3, 1, 0.707.
- 1.3** $R_2 = 250\text{ }\Omega$; the op-amp must source/sink the full 20 mA — near typical output-current limits.

Chapter 2 — The real Operational Amplifier

- 2.1** $A_{V,\min} = 990.1$, $A_{V,\max} = 998.0$ (spread 7.9), $\varepsilon_{A_V} \approx 0.8\%$.
- 2.2** $R_{in} \approx 400\text{ G}\Omega$, $R_{out} \approx 0.38\text{ m}\Omega$.
- 2.3** (a) $R_2 > 15\text{ k}\Omega$; e.g. $R_2 = 18\text{ k}\Omega$, $R_1 = 2\text{ k}\Omega$. (b) $R_2 > 150\text{ k}\Omega$ (bound scales as $1/I_{load}$).
- 2.4** $\text{CMRR} = 10^6$, so $A_C = A_d/\text{CMRR} = 1$.
- 2.5** (a) 5 V (current-limited). (b) 10 V (supply-limited). (c) $R_L^* = 400\text{ }\Omega$.
- 2.6** (a) Lower bound 24 k Ω , upper bound $\ll 2\text{ M}\Omega$; $R_2 = 100\text{ k}\Omega$ keeps a decade of margin both sides. (b) $v_{out,OFF} \approx 40 + 2 = 42\text{ mV}$ (offset *voltage* dominates).

Chapter 3 — Frequency response, stability and bandwidth

- 3.1** (a) $A_V = 10^{66/20} \approx 2000$ (just stable when $|T| = 0\text{ dB}$ at f_{p2}). (b) $A_V = 5000$: crossing near 400 kHz, below f_{p2} , so $\phi_M > 45^\circ$ — **stable**.
- 3.2** $M = 66\text{ dB}$; compensated $A_{d0} = 34\text{ dB}$ (≈ 50); $|T| = 0\text{ dB}$ at $f_{p2} = 1\text{ MHz}$ ($\phi_M = 45^\circ$), but the DC loop gain is now only 34 dB.
- 3.3** $f_{p1}^{new} = f_{p2}/T_0 = 10^6/10^3 = 1\text{ kHz}$; the full 60 dB DC loop gain survives.
- 3.4** $f_T = 10^6/25 = 40\text{ kHz} < 50\text{ kHz}$ — **fails**. Fixes: (i) faster part, $\text{GBWP} \geq 1.25\text{ MHz}$; (ii) cascade two $\times 5$ stages ($f'_T = 200\text{ kHz}$ each).
- 3.5** (a) $V_p \leq 7.96\text{ V}$. (b) $f \leq 7.96\text{ kHz}$, so a 10 V peak already slews at 10 kHz.

Chapter 4 — Applications

4.1 (a) $R_i'' = R_f/2$, $R_f/3$, $R_f/4$; add a +-side resistor $R_f/10$ to balance. (b) $R_1' = R_1'' = R_f/6$; add $R_2' = R_f$ to ground on the + side (a difference amplifier).

4.2 (a) $|A| = 1/(2\pi fRC) = 1.27$: output 127 mV at $+90^\circ$. (b) $dv/dt = I_{BIAS}/C = 10 \text{ V/s}$, so $\approx 1.2 \text{ s}$ to the rail.

4.3 $RC = 0.25 \text{ ms}$; e.g. $C = 10 \text{ nF}$, $R = 25 \text{ k}\Omega$.

Chapter 5 — The Operational Amplifier as a comparator

5.1 (a) $R_a = 15 \text{ k}\Omega$ (top), $R_b = 10 \text{ k}\Omega$ (bottom). (b) Out high for $v_{in} < 2.0 \text{ V}$, low for $v_{in} > 2.0 \text{ V}$. (c) Undefined for $1.95 \text{ V} < v_{in} < 2.05 \text{ V}$; need $\Delta V_S > 0.1 \text{ V}$.

5.2 (a) $V_{REF} = 0$, $R_2 = 5R_1$, e.g. $R_1 = 10 \text{ k}\Omega$, $R_2 = 47 \text{ k}\Omega$ (thresholds $\pm 2.1 \text{ V}$). (b) Tolerates noise up to $N_A \approx 2.1 \text{ V}$ peak.

5.3 (a) $V_{REF} = 0$, $R_2 = 4R_1$, e.g. $R_1 = 8.2 \text{ k}\Omega$, $R_2 = 33 \text{ k}\Omega$ (thresholds $\pm 2.98 \text{ V}$). (b) R_S adds to R_1 : thresholds widen to $\pm 3.35 \text{ V}$ (+12%) — buffer the source or keep $R_1 \gg R_S$.

5.4 $R_1 = R_2$ (e.g. both $10 \text{ k}\Omega$); $T = \tau \ln 9 = 2.20 \tau$, so $\tau = 45.5 \mu\text{s}$: $C = 1 \text{ nF}$, $R = 47 \text{ k}\Omega$ ($f \approx 9.7 \text{ kHz}$).

5.5 $R_2 = 4R_1$, e.g. $R_1 = 10 \text{ k}\Omega$, $R_2 = 39 \text{ k}\Omega$ ($V_{pp} \approx 5.1 \text{ V}$); $RC = 1 \text{ ms}$: $C = 10 \text{ nF}$, $R = 100 \text{ k}\Omega$.

Chapter 6 — Laboratory: Op-amp characterisation

6.1 The small step is *bandwidth-limited* ($t_{10-90} = 0.35/\text{BW}$, independent of step height); the large step is *slew-rate-limited* ($t_{10-90} = 0.8 V_H/\text{SR}$, which grows with the step height V_H).